

ACHYUT KATIYAR

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EDUCATION

Northeastern University

Master of Science in Computer Science

January 2025 – May 2027

Boston, Massachusetts, USA

SRM University AP

Bachelor of Technology in Computer Science and Engineering

October 2020 – May 2024

Andhra Pradesh, India

EXPERIENCE

Software Engineer Co-op

Maven Realty

June 2026 – December 2026

Cambridge, Massachusetts, USA

- Executed \$1.5M+ in leases through Aptava's automated DocuSign pipeline, covering 18 document types with signature and text tabs pulled from application data and signing order enforced for tenants, guarantors, landlords, and agents.
- Cleared intermittent production 504s on lease PDF generation by moving an S3 existence check to HeadObject, after unread GetObject bodies pinned one socket per draft against a 50-connection pool.
- Upgraded Aptava 9 Angular major versions to 18.2 and Node 16 to 24, then rewrote every broken Material component.
- Replaced an abandoned client-side pdfmake export with Puppeteer rendering an EJS template, then added a deploy hook that installs headless Chrome's shared libraries on Amazon Linux 2023 one at a time and logs failures instead of aborting.
- Built mavenrealty.com in Vue 3 and Vite with a URL codec that packs a filtered listing set into one shareable link, and tests that pin the old format so links already shared don't 404.

Teaching Assistant – CS2000

Northeastern University

September 2025 – April 2026

Boston, Massachusetts, USA

- Selected to TA CS2000 (Logic and Computation) across two consecutive semesters for a 500+ student introductory course.
- Held 4 hours of weekly office hours, graded 500+ assignments and exams, and wrote model-solution writeups on recursion, higher-order functions, and tree abstractions in Python and Pyret.

PROJECTS

Distributed Kanban Board | C++11, TCP, Vector Clocks, Primary-Backup Replication, Express, Socket.io, React

- Engineered a real-time collaborative Kanban in C++ with primary-backup replication. A hot standby promotes on a 5-second heartbeat and a full state transfer restores the rejoining master.
- Resolved concurrent edits with vector clocks and last-write-wins over a custom TCP binary protocol, fronted by an Express and Socket.io gateway. p90 held under 200ms and p99 under 500ms at 50 concurrent tasks.

MIT Bitcoin Expo 2026 Website | Next.js, TypeScript, Tailwind CSS, React Three Fiber, shadcn/ui

- Drew 100k+ visitors to the 500-attendee conference site with JSON-LD structured data and per-speaker landing pages.
- Cut the initial bundle by lazy-loading the React Three Fiber star field behind SSR-safe hydration boundaries.

PayPhone | Next.js, TypeScript, AWS, Terraform, Coinbase CDP, x402, Permit2, Anthropic API

- Built an AI agent that bills video calls by the second through Coinbase CDP x402, at \$0.01 per active second against one Permit2 witness capped at \$5. 6 settlements ran on-chain, 2 of them on Base mainnet.
- Hand-rolled the x402 verify-to-settle handshake instead of CDP's `paymentMiddleware` so settlement fires on the hangup webhook, and stood up Cognito, DynamoDB, and scoped IAM in one Terraform apply.

MIT Optimum Hacknet | Python, Bash, Streamlit, P2P Networking, Docker | Hackathon Winner

- Proved OptimumP2P holds a flat 161ms P95 while GossipSub degrades 140% at 20 msg/sec across a 12-node global cluster.
- Ran a parameter sweep that landed shard factor 8 and mesh degree 8–12 as the fastest config.

SkillBridge AI | Next.js, TypeScript, PostgreSQL, Prisma, Redis, OpenAI API, ElevenLabs, Docker

- Shipped a voice coaching PWA on GPT-4 and ElevenLabs, WCAG 2.1 AA compliant at a 95+ Lighthouse score.
- Cached GPT-4 completions in Redis to cut voice-chat latency, with Prisma Accelerate for Postgres connection pooling.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C++, C, SQL, Bash, Solidity
- **Frameworks:** Angular 18, Vue 3, React, Next.js, Node.js, Express, Spring Boot, RxJS, Prisma, Tailwind CSS
- **Cloud:** AWS (S3, Lambda, Elastic Beanstalk, CloudFront, DynamoDB, API Gateway, Cognito), Terraform
- **DevOps / Testing:** Docker, Kubernetes, Nginx, GitHub Actions, CI/CD, Vitest, Mocha, Puppeteer, Linux
- **Databases / APIs:** PostgreSQL, MongoDB, Redis, REST APIs, WebSockets, DocuSign, OAuth2, OpenAI

LEADERSHIP AND ACHIEVEMENTS

- **Hackathon Judge & Co-Chair — MIT Bitcoin Expo 2026:** Judged 47 teams alongside VCs, angel investors, and senior engineers at a 36-hour event with 150 participants and a \$20K prize pool, and designed the challenge tracks.
- **Hackathon Prizes (\$3,200+ across 6 wins):** \$1,500 Hacker House Goa 2024 | \$600 Move it with Aptos | \$500 Optimum Hacknet @ MIT | \$500 HackSRM 5.0 | \$100 Appathon 2022.